

Equivalence Partitioning (EP) Testing - BabePus

Equivalence Partitioning membagi input data menjadi kelompok-kelompok (partisi) yang seharusnya berperilaku sama. Jika satu nilai dalam partisi lulus testing, diasumsikan semua nilai dalam partisi yang sama juga lulus.

1. REGISTRATION FORM TESTING

Test Case: Email Input

Field: Email (campusEmail)

Partition	Range/Contoh	Expected	Test ID
Valid Email	user@campus.ac.id	Accept	EP-REG-E-001
	mahasiswa@univ.edu	Accept	EP-REG-E-002
	test.user+tag@kampus.org	Accept	EP-REG-E-003
Empty Email	(empty string)	Reject - Required	EP-REG-E-004
Invalid Format	user@.com	Reject - Invalid format	EP-REG-E-005
	user@@campus.ac.id	Reject - Invalid format	EP-REG-E-006
	user@campus	Reject - Invalid format (no TLD)	EP-REG-E-007
	user campus@.ac.id	Reject - Invalid format (space)	EP-REG-E-008
	(already exist di DB)	Reject - Email already used	EP-REG-E-009

Test Case:

TC-EP-REG-E-001: Valid Email

```
Input: user@campus.ac.id
Expected: Accept, proceed to next field
Result: PASS/FAIL

TC-EP-REG-E-004: Empty Email
Input: (empty)
Expected: Show error "Email is required"
Result: PASS/FAIL

TC-EP-REG-E-009: Duplicate Email
Input: existing@campus.ac.id
Expected: Show error "Email sudah terdaftar"
Result: PASS/FAIL
```

Test Case: Password Input

Field: Password

Partition	Range/Contoh	Expected	Test ID
Strong Password	Abc123!@#xyz	Accept	EP-REG-P-001
	SecurePass2024\$	Accept	EP-REG-P-002
Too Short	Pass1! (6 chars)	Reject - Min 8 chars	EP-REG-P-003
No Uppercase	password123!	Reject - Require uppercase	EP-REG-P-004
No Number	Password!@#	Reject - Require number	EP-REG-P-005
No Special Char	Password123	Reject - Require special char	EP-REG-P-006
Empty	(empty string)	Reject - Required	EP-REG-P-007
Space in Password	Pass 123!	Reject or Accept (tergantung requirement)	EP-REG-P-008

Test Case:

```
TC-EP-REG-P-001: Strong Password
Input: Abc123!@#xyz
Expected: Accept
Result: PASS/FAIL

TC-EP-REG-P-003: Password Too Short
Input: Pass1!
Expected: Show error "Minimal 8 karakter"
Result: PASS/FAIL

TC-EP-REG-P-006: No Special Character
Input: Password123
Expected: Show error "Harus mengandung karakter khusus"
Result: PASS/FAIL
```

Test Case: Student ID Input

Field: Student ID (Nomor Induk Mahasiswa)

Partition	Range/Contoh	Expected	Test ID
Valid Student ID	20220001 (8 digits)	Accept	EP-REG-S-001
	202200001 (9 digits)	Accept	EP-REG-S-002
Too Short	2022001 (7 digits)	Reject - Minimum 8 digits	EP-REG-S-003
Too Long	202200012345 (12 digits)	Reject - Maximum 10 digits	EP-REG-S-004
Contains Letters	2022A001	Reject - Numeric only	EP-REG-S-005
Empty	(empty)	Reject - Required	EP-REG-S-006
Special Characters	2022-0001	Reject - No special chars	EP-REG-S-007

Test Case:

```
TC-EP-REG-S-001: Valid Student ID
```

```
Input: 20220001
Expected: Accept
Result: PASS/FAIL
```

```
TC-EP-REG-S-003: Student ID Too Short
Input: 2022001
Expected: Show error "Minimal 8 digit"
Result: PASS/FAIL
```

```
TC-EP-REG-S-005: Contains Letters
Input: 2022A001
Expected: Show error "Hanya angka"
Result: PASS/FAIL
```

2. PRODUCT LISTING TESTING

Test Case: Price Input

Field: Product Price (Harga)

Partition	Range	Expected	Test ID
Valid Price	10,000 - 999,999,999 Rp	Accept	EP-PROD-PR-001
	50,000 Rp	Accept	EP-PROD-PR-002
	1,000,000 Rp	Accept	EP-PROD-PR-003
Price Too Low	< 10,000 Rp (e.g., 5,000)	Reject - Min 10K	EP-PROD-PR-004
Price Too High	> 999,999,999 Rp	Reject - Max exceeded	EP-PROD-PR-005
Zero/Negative	0	Reject - Must > 0	EP-PROD-PR-006
	-50,000	Reject - Cannot be negative	EP-PROD-PR-007
Empty	(empty)	Reject - Required	EP-PROD-PR-008
Non-Numeric	"lima ribu"	Reject - Numeric only	EP-PROD-PR-009

Partition	Range	Expected	Test ID
Decimal	50,000.50	Accept or Reject (tergantikan requirement)	EP-PROD-PR-010

Test Case:

```
TC-EP-PROD-PR-001: Valid Price
Input: 50000
Expected: Accept
Result: PASS/FAIL

TC-EP-PROD-PR-004: Price Too Low
Input: 5000
Expected: Show error "Harga minimum Rp 10.000"
Result: PASS/FAIL

TC-EP-PROD-PR-006: Zero Price
Input: 0
Expected: Show error "Harga harus lebih dari 0"
Result: PASS/FAIL
```

Test Case: Product Name

Field: Product Name (Nama Produk)

Partition	Range	Expected	Test ID
Valid Name	3-100 chars, alphanumeric + space	Accept	EP-PROD-N-001
	"Laptop ASUS"	Accept	EP-PROD-N-002
	"HP Bekas - Mulus Semua"	Accept	EP-PROD-N-003
Too Short	1-2 chars (e.g., "HP")	Reject - Min 3 chars	EP-PROD-N-004
Too Long	> 100 chars	Reject - Max 100 chars	EP-PROD-N-005
Empty	(empty)	Reject - Required	EP-PROD-N-006

Partition	Range	Expected	Test ID
Special Chars	"Laptop @#\$\$%"	Accept or Reject (depending on rules)	EP-PROD-N-007
Numbers Only	"12345"	Accept (should be allowed)	EP-PROD-N-008

Test Case:

```

TC-EP-PROD-N-001: Valid Product Name
Input: Laptop ASUS ROG Gaming
Expected: Accept
Result: PASS/FAIL

TC-EP-PROD-N-004: Product Name Too Short
Input: HP
Expected: Show error "Minimal 3 karakter"
Result: PASS/FAIL

TC-EP-PROD-N-005: Product Name Too Long
Input: (101 characters)
Expected: Show error "Maksimal 100 karakter"
Result: PASS/FAIL

```

3. OFFER/PRICE NEGOTIATION TESTING

Test Case: Offered Price

Field: Offered Price (Harga yang ditawarkan)

Partition	Kondisi	Expected	Test ID
Valid Offer	< Original Price	Accept	EP-OFFER-OP-001
	Original: 100K, Offer: 80K	Accept	EP-OFFER-OP-002
Equal to Original	Offered Price = Original Price	Accept (tapi tidak ada negotiation)	EP-OFFER-OP-003
Higher than Original	Offered Price > Original Price	Reject or Warning	EP-OFFER-OP-004

Partition	Kondisi	Expected	Test ID
	Original: 100K, Offer: 120K	Reject - Cannot exceed ask price	EP-OFFER-OP-005
Negative or Zero	≤ 0	Reject - Must > 0	EP-OFFER-OP-006
Too Low	$< 10\%$ dari original price (e.g., Original: 100K, Offer: 5K)	Warning or Accept?	EP-OFFER-OP-007
Empty	(empty)	Reject - Required	EP-OFFER-OP-008

Test Case:

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TC-EP-OFFER-OP-001: Valid Offer (Lower than Original)
Input: Original Price: 100000, Offered Price: 80000
Expected: Accept, show success message
Result: PASS/FAIL

TC-EP-OFFER-OP-004: Offered Higher than Original
Input: Original Price: 100000, Offered Price: 120000
Expected: Show warning "Harga penawaran tidak boleh melebihi harga
ask"
Result: PASS/FAIL

TC-EP-OFFER-OP-007: Extremely Low Offer
Input: Original Price: 100000, Offered Price: 5000
Expected: Accept (or show warning)
Result: PASS/FAIL

```

4. RATING/REVIEW TESTING

Test Case: Review Rating

Field: Star Rating (1-5)

Partition	Range	Expected	Test ID
Valid Rating	1 star	Accept	EP-REVIEW-R-001

Partition	Range	Expected	Test ID
	2 stars	Accept	EP-REVIEW-R-002
	3 stars	Accept	EP-REVIEW-R-003
	4 stars	Accept	EP-REVIEW-R-004
	5 stars	Accept	EP-REVIEW-R-005
Below Range	0 stars	Reject - Minimum 1 star	EP-REVIEW-R-006
	-1 star	Reject - Invalid	EP-REVIEW-R-007
Above Range	6 stars	Reject - Maximum 5 stars	EP-REVIEW-R-008
	10 stars	Reject - Invalid	EP-REVIEW-R-009
Non-Numeric	"Good"	Reject - Numeric only	EP-REVIEW-R-010
Empty	(no selection)	Reject - Required	EP-REVIEW-R-011

Test Case:

```

TC-EP-REVIEW-R-005: Valid 5-Star Rating
Input: 5
Expected: Accept, display filled 5 stars
Result: PASS/FAIL

TC-EP-REVIEW-R-006: Zero Rating
Input: 0
Expected: Show error "Minimal 1 bintang"
Result: PASS/FAIL

TC-EP-REVIEW-R-008: 6-Star Rating
Input: 6
Expected: Show error "Maksimal 5 bintang"
Result: PASS/FAIL

```

Test Case: Review Comment

Field: Review Comment (Komentar)

Partition	Range	Expected	Test ID
Valid Comment	10-500 chars	Accept	EP-REVIEW-C-001
	"Barang bagus, cepat sampai"	Accept	EP-REVIEW-C-002
Too Short	< 10 chars (e.g., "Bagus!")	Reject - Min 10 chars	EP-REVIEW-C-003
Too Long	> 500 chars	Reject - Max 500 chars	EP-REVIEW-C-004
Empty	(optional field)	Accept	EP-REVIEW-C-005
Special Characters	"Great! @#\$%"	Accept	EP-REVIEW-C-006
Numbers Only	"1234567890"	Accept	EP-REVIEW-C-007

Test Case:

```
TC-EP-REVIEW-C-002: Valid Comment
Input: Barang bagus sekali, seller responsif, dan pengiriman cepat
Expected: Accept
Result: PASS/FAIL
```

```
TC-EP-REVIEW-C-003: Comment Too Short
Input: Bagus!
Expected: Show error "Minimal 10 karakter"
Result: PASS/FAIL
```

```
TC-EP-REVIEW-C-004: Comment Too Long
Input: (501+ characters)
Expected: Show error "Maksimal 500 karakter"
Result: PASS/FAIL
```

5. LOGIN TESTING

Test Case: Email (Login)

Field: Email (Login)

Partition	Contoh	Expected	Test ID
Valid Email Format	registered@campus.ac.id	Try to authenticate	EP-LOGIN-E-001
Valid Format, Not Registered	notregistered@campus.ac.id	Reject - Email not found	EP-LOGIN-E-002
Empty Email	(empty)	Reject - Required	EP-LOGIN-E-003
Invalid Format	user@.com	Reject - Invalid format	EP-LOGIN-E-004
Case Variation	USER@CAMPUS.AC.ID	Accept (case-insensitive)	EP-LOGIN-E-005

Test Case:

```
TC-EP-LOGIN-E-001: Valid Email Registered
Input: user@campus.ac.id (with correct password)
Expected: Login success
Result: PASS/FAIL

TC-EP-LOGIN-E-002: Valid Format But Not Registered
Input: notregistered@campus.ac.id
Expected: Show error "Email tidak ditemukan"
Result: PASS/FAIL

TC-EP-LOGIN-E-003: Empty Email
Input: (empty)
Expected: Show error "Email is required"
Result: PASS/FAIL
```

Test Case: Password (Login)

Field: Password (Login)

Partition	Contoh	Expected	Test ID
Correct Password	(matching email password)	Accept	EP-LOGIN-P-001
Wrong Password	(not matching)	Reject - Invalid credentials	EP-LOGIN-P-002
Empty Password	(empty)	Reject - Required	EP-LOGIN-P-003
Too Short	Pass1	Reject or Accept based on validation	EP-LOGIN-P-004
Case Sensitive	CORRECTPASS (if correct is lowercase)	Reject - Password incorrect	EP-LOGIN-P-005

Test Case:

```

TC-EP-LOGIN-P-001: Correct Password
Input: (correct password for registered email)
Expected: Login success
Result: PASS/FAIL

TC-EP-LOGIN-P-002: Wrong Password
Input: wrongpassword123
Expected: Show error "Email atau password salah"
Result: PASS/FAIL

TC-EP-LOGIN-P-003: Empty Password
Input: (empty)
Expected: Show error "Password is required"
Result: PASS/FAIL

```

Summary Tabel EP Test Cases

Module	Partitions	Total Test Cases	Coverage
Registration	Email: 9	27	Comprehensive
	Password: 8		
	Student ID: 7		
Product Listing	Price: 10	19	Comprehensive

Module	Partitions	Total Test Cases	Coverage
	Product Name: 8		
Offer	Offered Price: 8	8	Good
Review	Rating: 11	19	Comprehensive
	Comment: 7		
Login	Email: 5	10	Good
	Password: 5		
TOTAL		83	Good Coverage

Best Practices untuk EP Testing

1. Identifikasi Partition dengan Clear Boundaries

- Pisahkan input valid dan invalid dengan jelas

2. Test Representative dari Setiap Partition

- Ambil 1 nilai per partition (tidak perlu semua)

3. Include Boundary Values

- Minimum, maximum, one below, one above

4. Dokumentasi yang Jelas

- Catat partition definitions dan test cases

5. Expected vs Actual

- Bandingkan hasil actual dengan expected result